

and serving features, whereas ZenML assists teams in creating reusable pipeline components.

The transparency that open source software provides is among its most significant advantages. Teams can examine the real source code, comprehend how it works, and alter it to suit the requirements of their projects. Even when team members have different levels of technical expertise, this transparency fosters trust and facilitates easier collaboration.

Open source MLOps is now much more than just a way to cut costs. It has emerged as a potent force for innovation, dismantling the obstacles of closed platforms and costly infrastructure. Open source technologies are enabling teams, from small startups to huge corporations and academicians, to create scalable, production-ready machine learning pipelines with more control and flexibility.

Conceptual workflow of ML project with open source MLOps

Building a model is only one aspect of managing a machine learning project. It entails several related actions that need to be meticulously planned and monitored. Open source MLOps tools give this process structure by outlining precise, scalable, and repeatable routines. Data collection is frequently the first step in the process. Raw datasets are collected, labelled, and kept here. These datasets are versioned with the aid of tools such as DVC, which enable teams to revert to an earlier version when necessary. When models later need to be retrained or audited, this versioning is essential. Preprocessing is the next step. This covers feature extraction, transformation, and cleaning. To manage feature stores that supply consistent features during both training and inference, some teams employ technologies like Feast.

As models are built and trained, experiment tracking becomes essential. MLflow allows teams to log

Case study: How open source MLOps tools helped in COVID-19 diagnosis workflow management

Background: Healthcare systems around the world experienced an overwhelming influx in data during the COVID-19 epidemic, necessitating the quick creation and implementation of machine learning models to help with patient triage, disease diagnosis, and prediction. However, irregular processes, versioning problems, and a lack of scalable infrastructure hindered the practical application of these AI models for most healthcare teams.

Problem statement: To detect COVID-19 positive cases, a top government health research institute sought to use machine learning models to evaluate patient vitals and chest X-ray data. Ad hoc pipelines and manual procedures caused the deployment and update cycles to be sluggish and unreliable, even when promising model accuracy was attained in research labs. Delays in interventions and ineffective use of resources resulted from the lack of a standardised way to manage trials, version datasets, or monitor models in real-time.

Proposed solution: The organisation used open source MLOps tools to handle the entire machine learning pipeline to address this difficulty.

- MLflow was used to track experiments and manage model versions.
- DVC handled data versioning and ensured reproducibility across clinical trial phases.
- Kubeflow automated the training and deployment pipeline, allowing quick iteration and retraining based on incoming data.
- Prometheus and Grafana enabled real-time monitoring of deployed models in hospitals.

The ML workflow was made more efficient by using these tools together, which covered raw data collection, preprocessing, model training, evaluation, and deployment under a monitored, scalable infrastructure.

Outcome: The time-to-deployment was drastically shortened from weeks to days with the help of MLOps. With updated models, the clinical teams could react to new variants more quickly, and model accuracy was constantly checked for declines in performance. Reproducible tests also produced more reliable medical results and clinical deployment authorisation.

This is how mission-critical health projects can achieve operational excellence using open source MLOps technologies, especially in the face of limited resources.

Key takeaways

- MLOps is essential in high-stakes domains like healthcare, where lives depend on consistent and rapid model deployment.
- Open source tools reduce dependency on expensive platforms and empower public institutions.
- Streamlined workflows improve not just operational efficiency, but also public trust and decision-making.

Source: NCBI COVID-19 AI Diagnosis Study (<https://pmc.ncbi.nlm.nih.gov/articles/PMC7395236/>)

parameters, results and artifacts for each experiment. This visibility helps in comparing approaches and choosing the most effective one. The next phase is pipeline orchestration. Airflow and similar tools let teams schedule and automate workflows. From fetching new data to retraining models on a schedule, these tools ensure the

process runs smoothly without manual effort. Deployment and monitoring are the final pieces. MLOps ensures that models move into production in a controlled, traceable way. Teams can monitor performance over time and trigger retraining when needed. Each tool in this pipeline plays a specific role, but together they create a reliable